Digital Technology: Programmable Logic Devices (PLD)

EIT Bachelor - Study Guide

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1 Internal Structure of Simple PLDs (GAL)

Simple Programmable Logic Devices (SPLDs), specifically the **GAL (Generic Array Logic)**, are organized into specific logic planes[cite: 104, 110]:

- **AND-Plane (Programmable):** Inputs are connected to a grid where connections are programmed to create product terms.
- **OR-Plane (Fixed):** These terms are fed into an OR-gate to produce the final logic sum.
- **OLMC (Output Logic Macrocell):** Contains flip-flops to allow sequential logic like state machines[cite: 66, 104].

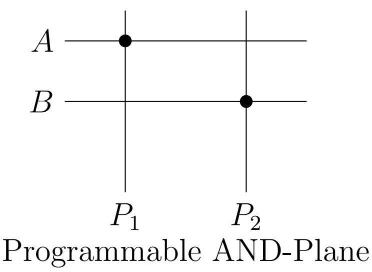

Figure 1: Simplified internal organization of an SPLD logic plane.

2 Difference between Simple and Complex PLD (CPLD)

The difference lies in the scale and the way components are connected:

- **Integration**: A CPLD consists of multiple SPLD-like function blocks on a single chip.
- **Interconnect**: CPLDs use a central **Programmable Interconnect Matrix (PIM)** to link these blocks, whereas SPLDs are self-contained[cite: 61, 67].

3 Structure of Field-Programmable Gate Arrays (FPGA)

FPGAs use a distributed "Sea of Gates" architecture[cite: 122]:

- **Configurable Logic Blocks (CLB):** Arranged in a grid, these contain Look-Up Tables (LUT) and Flip-Flops[cite: 60, 66].
- **Programmable Routing:** A grid of wires and switches allows connections between any CLBs[cite: 67].

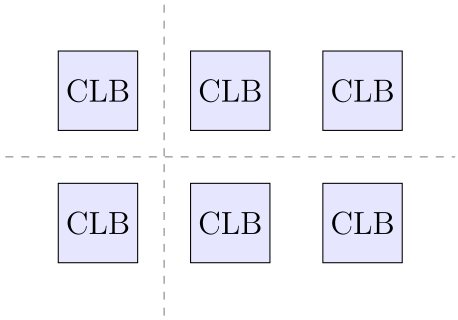

Figure 2: FPGA grid structure with routing channels.

4 Comparison: CPLD vs. FPGA

The differences regarding structure, technology, and programming are as follows:

Feature	**CPLD**	**FPGA**
Structure	Centralized Matrix [cite: 65]	Distributed Grid [cite: 122]
Technology	Flash / EEPROM [cite: 119]	SRAM [cite: 122]
Programming	Non-volatile (Instant-on) [cite: 119]	Volatile (Needs reload) [cite: 122]

Table 1: Comparison of CPLD and FPGA technologies.